

Trainings ▾

Preparatory Steps

QSK45 and QSK60 Single Stage

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the turbochargers. Refer to Procedure 010-035 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)

QSK60 Marine Applications

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Remove the turbochargers. Refer to Procedure 010-035 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)
- Remove the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11
(/qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)

QSK60 Two Stage Industrial and Power Generation

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the low-pressure turbochargers. Refer to Procedure 010-035 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)
- Remove the two-stage turbocharger frame. Refer to Procedure 010-082 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html)
- Remove the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)

Remove

The QSK45 engine has three oil cooler covers and the QSK60 single- and two-stage engines have four oil cooler covers. All covers are removed the same way. Each cover contains two

threaded holes that can be used to aid in the removal of the cover. **Only** the QSK60 two-stage engine has frame support brackets welded on the two center covers.

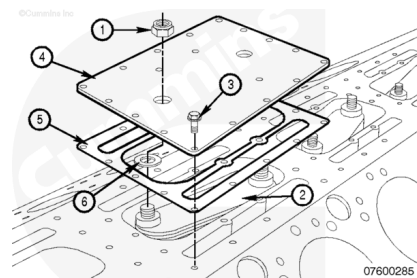
Remove two locknuts (1) from the oil cooler element mounting studs (2).

Remove all twenty capscrews (3).

Install two capscrews (3) in the threaded holes of the cover (4).

Thread the capscrews in evenly to remove the cover.

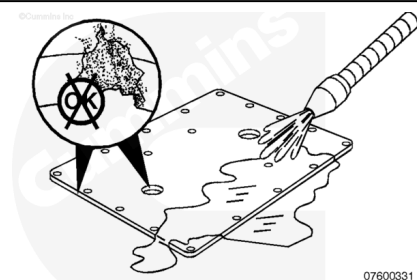
Remove the cover (4) and discard the gaskets (5 and 6).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the covers with solvent and dry with compressed air.

Inspect the covers for corrosion and other damage.

Replace the covers if corroded or otherwise damaged.

Install

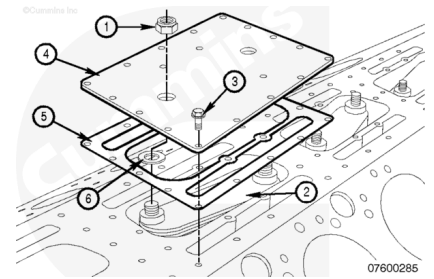
The QSK45 engine has three oil cooler covers, and the QSK60 single- and two-stage engines have four oil cooler covers. The rear two-stage frame support bracket **must** be installed after

the rear cover is installed. **Only** the QSK60 two-stage engine has frame support brackets welded on the two center covers.

Install one edge-molded gasket (6) on each of the oil cooler element mounting studs (2). Use Loctite® 271, Part Number 3375068, sparingly on the oil cooler stud threads.

Install a new gasket (5) and the oil cooler cover (4). Position twenty capscrews (3) to secure the cover in position, but do **not** tighten the capscrews.

Install the header plate gaskets with the part number on the gasket facing upwards over the guide studs and making sure all mounting holes line up. Position the twenty capscrews (3) to secure the cover in position, but do **not** tighten the capscrews.



Tighten the oil cooler cover capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

Tighten the self-locking nuts (21 and 22).

Torque Value:

Initial Torque Value 45 n•m [33 ft-lb]

Torque Value:

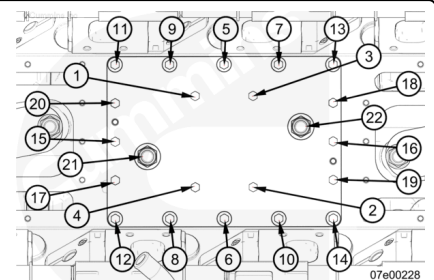
Final Torque Value 95 n•m [70 ft-lb]

For industrial engines, tighten the two-stage frame support bracket capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

For four turbocharger engines, tighten the exhaust collector support bracket capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

QSK45 and QSK60 Single Stage

- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Install the turbochargers. Refer to Procedure 010-034 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html)
- Fill the cooling system Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

- Operate the engine to normal operating temperature. Check for leaks.

QSK60 Marine Applications

- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Install the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html)
- Install the turbochargers. Refer to Procedure 010-034 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html)
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Fill the cooling system Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to normal operating temperature. Check for leaks.

QSK60 Two Stage Industrial and Power Generation

- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Install the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html)
- Install the two-stage turbocharger frame. Refer to Procedure 010-082 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html)
- Install the low-pressure turbochargers. Refer to Procedure 010-035 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-035-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to normal operating temperature. Check for leaks.